Moussa Traore

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Software Engineer: Full Stack, Data Pipelines and Reporting

Software Engineer with 2 years of experience building full-stack web applications and data platforms in Python, Java and JavaScript. Built Airflow pipelines ingesting 10M+ records, cut analytical query times by up to 60% and developed Java EE and Angular web services for a national data platform.

Skills

- **Programming Languages:** Python | Java | SQL | JavaScript | TypeScript | Bash | PHP | C++ (basics)
- **Backend and Data:** FastAPI | REST APIs | Java EE | PostgreSQL | MySQL | Oracle | Apache Airflow | Apache Spark | Hadoop
- **Frontend:** Angular | React | HTML | CSS | Vite | Tailwind CSS
- **DevOps and Build:** GitLab CI/CD | Docker | Docker Compose | Linux | Git | Traefik | Package Managers (pip, npm)
- **Security:** Identity and Access Management (Keycloak, OpenID Connect) | Vulnerability Management (Trivy)
- **Computer Science:** Data Structures and Algorithms
- **Languages:** French (native) | Bambara (native) | English (professional working proficiency)

Work Experience

French National Centre for Scientific Research (CNRS) Besançon, France
Software Engineer, Data Platform *02/2025 – Present*

- **Data pipelines:** Built and validated Apache Airflow pipelines in Python ingesting 10M+ environmental sensor records from several sources.
- **Performance:** Rewrote and indexed PostgreSQL and MySQL queries, cutting response times of key analytical queries by up to 60%.
- **Developer tooling:** Created internal web tools for research teams in Python, PHP and JavaScript.
- **Release engineering:** Shipped releases with Docker Compose through a tag-based GitLab CI/CD workflow.
- **Identity and access management:** Operated Keycloak (OpenID Connect) authentication for 10 open-source services (FastAPI, Grafana, Superset, Nextcloud) on Linux behind Traefik.
- **Vulnerability management:** Added Trivy container scanning as a blocking GitLab CI/CD stage, then triaged and fixed reported CVEs before each release to keep vulnerable images out of production.

Atlantic Wetlands Forum Rochefort, France
Software Engineer *08/2024 – 01/2025*

- **Software quality:** Introduced unit testing and verification-driven deployment, reducing production bugs by 25% and deployment time by 40%.
- **Backend:** Developed Java EE back-end features and web services with an Angular front end for a national environmental data platform, improving page-load times and scalability by 35%.

Atlantic Wetlands Forum Rochefort, France
Software Engineering Intern *03/2024 – 08/2024*

- **Requirements:** Turned stakeholder needs into 15 functional and technical requirements for a national Oracle wetlands database.
- **Architecture:** Wrote the technical specification of a data architecture connecting 50,000+ datasets across systems, improving information retrieval by 40%.

Education

MSc in Computer Science, Databases and Artificial Intelligence 2024
Université de Bourgogne, Dijon, France
BSc in Mathematics and Computer Science 2022
Université de Bourgogne, Dijon, France

Projects

- **AI Mali Platform, Contributor | 2025 – Present | aimali.ai:** Built a bilingual (FR/EN) website as one of 3 contributors to its monorepo: React, Vite and Tailwind public site and admin CMS, REST API on Cloudflare Workers (Hono) with Neon Postgres, R2 and KV, deployed on Cloudflare Pages.
- **AI Mali, Early Member | 2025 – Present | aimali.ai:** Researched distributed AI for language modeling; collected and annotated 10,000+ Bambara words to build a dataset for a low-resource language.
- **TimaTon (AI Mali), Contributor | 2025 – Present:** Contributed to an offline-first reforestation app for field agents in Mali: React Native mobile app, Node.js TypeScript API on PostgreSQL and a React back-office.

Awards and Accolades

- **LeetCode | 221 problems solved (121 medium, 18 hard), mostly in Python** | 200 Days Badge 2026 | longest daily streak of 176 days